

ETL INTRODUCTION

E-EXTRACT

T-TRANSFORM

L-LOAD

ETL is a data integration process used to collect data from different sources ,transform it into a usable format and load it into a target system such as a data warehouse or database.

ETL TOOLS:

Open source

Hadoop

Scoop

ETL combines all this data into one centralised system for analytics

Data injection tool :- tools used to inject data

For offline data migration we need ETL tools and for online we need streaming tool.

1.ETL Architecture

Source system ->ETL process ->Data warehouse ->reporting tools

Example:

Csv file

Database

API

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Extract

|

Transform

|

Load

|

Datawarehouse

|

Dashboard/BI Tools

2.Data extraction

Data extraction means collecting data from different sources

Realational database(my sql,sql server)

,flat files(csv,excel),

APIs,

cloud storage,

3.Data Transformation

Transformation means cleaning and modifying data to make it usable.

Common transformation tasks

-removing duplicates

-data cleaning

-format conversion

- aggregation
- filtering
- data validation

4.Data Loading

Loading means moving processed data into a target system

Target system can include:

- data warehousing
- database
- datalake
- bi tools

Sql loading exp: insert into sale_warehouse

Select * from cleaned_sales;

Python loading exp: data.to_sql("sales_table",connection)

Types of loading

1.full load

Entire data loaded.

Exp:delete old data and load entire new dataset

Only new data added

Exp:load sales after yesterday

5.ETL TOOLS and Technologies

Etl can be implemented using many tools.

Tool	description
Informatica	enterprise ETL platform
Talend	open-source ETL tool
Pentaho	data integration tool

Modern cloud ETL tools.

Tool	platform
Azure data factory	microsoft azure
Aws glue	amazon
Google dataflow	google cloud

Big data ETL Tools.

Tool	platform
Apache spark	large scale processing
Apache airflow	workflow orchestration
Kafka	streaming data pipelines

Example ETL pipeline

Source -> ETL Tools -> Data warehouse

Example using azure

Azure blob storage

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Azure data factory

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Transformation

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Azure synapse

ASSIGNMENT

Students perform a simple etl pipeline

Dataset

Sales_data.csv

Orderid,Product,Price

1,laptop,500000

2,phone,200000

3,Tablet,300000

Tasks

Step1 Extract data from csv.

Step 2 Transform:

Remove null values

Calculate tax(price *1.18)

Step3 Load into sql database.

Practice questions:

1.what is ETL?

2.what are the three stages of ETL?

3,What is data transformation?

4..difference between full load and incremental load

5.Name any two ETL tools?